

Three Kobon numbers, a crossing-refined capacity theorem, and the exact status of $K(14)$

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Abstract

The Kobon triangle problem asks for the maximum number of non-overlapping triangles determined by n straight lines. We first separate three quantities that the literature conflates: $a_3^s(n)$, triangular faces of *simple* arrangements; $K(n)$, triangular faces of *arbitrary* arrangements (the object of Clément–Bader’s definition, in which a counted triangle is “realized by 3 straight line segments”); and $K_{\text{gen}}(n)$, the maximum size of a family of triangles with pairwise disjoint interiors in which other lines *may* cross a counted triangle. Then $a_3^s \leq K \leq K_{\text{gen}}$, and we show that the classical segment-counting arguments bound only the first two.

We prove a capacity inequality valid for K_{gen} with arbitrary parallels and multiple points,

$$3|S| + C + 2Q + \sum_p k_p(k_p - 4) \leq n(n - 2),$$

where C counts line/triangle interior incidences, together with an exact bounded-face budget; combining them gives the closed form $K_{\text{gen}}(n) \leq \lfloor (n - 2)(5n - 3)/12 \rfloor$, in which triple points are exactly neutral. To our knowledge these are the only bounds proved for K_{gen} .

We then use exact SAT certificates to settle small cases in this strongest convention: $K_{\text{gen}}(n)$ for every $n \leq 10$ — with DRAT-verified refutations at $n = 4, \dots, 9$ — which are strictly stronger statements than the corresponding classical ones because they also exclude crossed families. On the constructive side we independently verify, in exact rational arithmetic and with three implementations, a 14-line arrangement with 54 pairwise interior-disjoint triangles, so $K_{\text{gen}}(14) \geq 54$; and we prove that every currently known extremal arrangement at $n = 11, 14, 18, 20$ attains its published count as its exact optimum, so crossings never help inside known configurations. Finally we observe that the entry $a(14) = 54$ recorded as exact in OEIS A006066 on 21 August 2026 pairs a non-simple construction with a simple-arrangement upper bound; the rigorous status is $54 \leq K(14) \leq 55$, and we describe the decision procedure that closes it, together with two equality-branch obstruction theorems ($n = 14$ and $n = 18$, machine-checked) and a proof that the method cannot close the analogous branch at $n = 20$.

1 Three quantities

Let $\mathcal{A}$ be an arrangement of $n \geq 3$ distinct lines in the real affine plane, not all parallel (*essential*). A *segment* of $\mathcal{A}$ is a bounded maximal piece of a line between two consecutive crossing points.

Definition 1. (i) $a_3^s(n)$: the maximum number of triangular bounded faces of a *simple* arrangement of n lines (no two parallel, no three concurrent).

(ii) $K(n)$: the maximum, over all arrangements of n lines, of the number of triangles each of whose three sides is a single segment of $\mathcal{A}$, no two of them overlapping. Equivalently — since

a line entering the interior of a triangle must cross two of its sides and thereby subdivide them — the maximum number of triangular bounded faces of an arbitrary arrangement.

- (iii) $K_{\text{gen}}(n)$: the maximum, over all arrangements, of the size of a family S of triangles bounded by triples of lines of $\mathcal{A}$ whose open interiors are pairwise disjoint. Members of S may be crossed by other lines of $\mathcal{A}$.

Clearly $a_3^s(n) \leq K(n) \leq K_{\text{gen}}(n)$.

Clément and Bader [3] define a Kobon triangle as “a triangle that is realized by 3 straight lines segments and which does not overlap with other triangles”, i.e. they work with K ; their Lemma 1 argument enumerates the cases “side shared by two triangles” (which forces concurrency at an endpoint) and “segment used by no triangle”, and their mod-6 improvement passes through *perfect configurations*, defined as arrangements of *pairwise intersecting* lines in which every segment is the side of exactly one triangle, and shown in their Lemma 2 to have no three concurrent lines. Their bound

$$K(n) \leq \lfloor n(n-2)/3 \rfloor - \mathbf{1}_{n \bmod 6 \in \{0,2\}} \quad (1)$$

is therefore a statement about K ; the case of a counted triangle whose sides are subdivided by crossings is not part of their case analysis, and their claim that a segment serving two triangles forces a multiple point at one of its endpoints fails once subdivided sides are admitted. Bartholdi, Blanc and Loisel [1] and Blanc [2] bound a_3^s . Thus:

The base segment bound $\lfloor n(n-2)/3 \rfloor$ is established for K including non-simple arrangements by Felsner and Kriegel [4] (so used, e.g., in [6]); the mod-6 improvement (1) for K rests on the unpublished draft [3]; the sharpest numerical bounds [1, 2] are proved for a_3^s only and hence do not apply to non-simple record arrangements; and no published bound is proved for K_{gen} .

This has an immediate consequence for the current record. On 21 August 2026 OEIS A006066 [9] was updated to list $a(14) = 54$ as exact, attributed to Maiorana’s arrangement [8]. That arrangement is *not* simple: it has two triple points (on lines $\{0, 4, 10\}$ and $\{0, 11, 12\}$, sharing line 0) and no parallels. The matching upper bound 54 in the same table is the BBL value $\lfloor n(n-7/3)/3 \rfloor$, proved for simple arrangements. For K , the peer-reviewed non-simple segment bound [4] gives 56 at $n = 14$, and the mod-6 improvement (1) of the draft [3] gives 55. Hence:

Proposition 2. *With the published literature, $54 \leq K(14) \leq 56$ unconditionally, and $K(14) \leq 55$ if the draft [3] is granted; for the strongest convention $54 \leq K_{\text{gen}}(14) \leq 67$. In every case $a(14) = 54$ is a lower bound paired with a simple-arrangement upper bound, so its exactness is not established.*

Contributions.

1. A capacity theorem for K_{gen} (Theorem 5), with exact degeneracy penalty and crossing charge, and a closed-form ceiling (Theorem 9).
2. A certified initial segment in the strongest convention: $K_{\text{gen}}(4), \dots, K_{\text{gen}}(9) = 2, 5, 7, 11, 15, 21$, each with a **drat-trim**-verified refutation of $K_{\text{gen}}(n) + 1$, plus $K_{\text{gen}}(10) = 25$ from a prior cube cover (Theorem 11).
3. An independently verified $K_{\text{gen}}(14) \geq 54$ (Theorem 12) and an exact optimality census of all known extremal arrangements (Theorem 13), which shows $K = K_{\text{gen}}$ on each of them.

4. Two equality-branch obstruction theorems with machine-checked finite certificates (Theorems 14, 15) and a proof that the same method cannot close the analogous branch at $n = 20$ (Proposition 16).
5. The status audit of $K(14)$ (Proposition 2) and the decision procedure that closes it.

2 Notation

For $\ell \in \mathcal{A}$ let q_ℓ be the number of lines parallel to ℓ and $Q = \frac{1}{2} \sum_\ell q_\ell$. A *multipoint* has multiplicity $k_p \geq 3$; N_k counts points of multiplicity k . On ℓ , r_ℓ is the number of distinct crossings, a_ℓ the number of multipoints, and $m_\ell = r_\ell - 1$ the number of bounded elementary gaps. For an admissible family S (Definition 1(iii)) set

$$\sigma_\ell = \#\{T \in S : \ell \text{ supports a side of } T\}, \quad \gamma_\ell = \#\{T \in S : \ell \cap \text{int}(T) \neq \emptyset\},$$

so $\sum_\ell \sigma_\ell = 3|S|$ and $C := \sum_\ell \gamma_\ell$ is the total number of line/triangle interior incidences. Write $T = |S|$ and $\Delta = n(n-2) - 3T$.

3 The capacity theorem

Lemma 3 (incidence identity). $m_\ell + 2a_\ell = n - 2 - q_\ell - \sum_{p \ni \ell} (k_p - 4)$ for every $\ell \in \mathcal{A}$.

Proof. The $n - 1 - q_\ell$ lines meeting ℓ do so in r_ℓ distinct points, and a multipoint p on ℓ absorbs $k_p - 1$ of them into one point, so $n - 1 - q_\ell = r_\ell + \sum_{p \ni \ell} (k_p - 2)$. Hence $m_\ell + 2a_\ell = r_\ell - 1 + 2a_\ell = n - 2 - q_\ell - \sum_{p \ni \ell} (k_p - 2) + 2a_\ell$, and a_ℓ is the number of summands. $\square$

Lemma 4 (per-line capacity). $\sigma_\ell + \gamma_\ell \leq m_\ell + 2a_\ell$ for every $\ell \in \mathcal{A}$ and every admissible S .

Proof sketch. Each $T \in S$ with a side on ℓ determines the side interval $I_T \subset \ell$; each T crossed by ℓ determines the chord interval $J_T = \ell \cap \text{int}(T)$. Disjointness of interiors makes the J_T pairwise disjoint, makes each J_T disjoint from the interiors of side intervals on the same side of ℓ , and makes side intervals on a fixed side of ℓ interior-disjoint. Every such interval contains at least one elementary gap, and two intervals can claim the same gap only when they lie on opposite sides of ℓ and abut a common multipoint of ℓ , each of which admits at most two coincidences. Charging one gap per interval and two tokens per multipoint yields the claim; the exhaustive case analysis is carried out in the companion document [10, §3] and was re-derived by an adversarial audit over 3,016 arrangements and 812,486 admissible families with 5,900,863 exact linewise checks. $\square$

Theorem 5 (capacity). *For every essential arrangement of n distinct affine lines and every admissible family S ,*

$$3|S| + C + 2Q + \sum_p k_p(k_p - 4) \leq n(n - 2). \quad (2)$$

Proof. Sum Lemma 4 over all lines and substitute Lemma 3; the left side gives $3|S| + C$, and on the right $\sum_\ell q_\ell = 2Q$ while each multipoint lies on exactly k_p lines, so $\sum_\ell \sum_{p \ni \ell} (k_p - 4) = \sum_p k_p(k_p - 4)$. $\square$

Setting $C = 0$ recovers, with explicit degeneracy corrections, the classical segment bound for K ; the point of (2) is that the crossing charge makes it valid for K_{gen} .

Remark 6. Neither correction term can be replaced by the naive alternatives: the variants with $\binom{k_p-2}{2}$ or $(k_p-2)^2$ in place of $k_p(k_p-4)$ are false, with exact counterexamples in [10, §6]. The all-parallel class is the unique inessential exception.

Corollary 7. *If $\mathcal{A}$ has no triple point then $|S| \leq \frac{1}{3}(n(n-2)-C-2Q-\sum_{k \geq 4} N_k k(k-4)) \leq n(n-2)/3$. In general $|S| \leq n(n-2)/3 + N_3 - \frac{1}{3}(C+2Q+\sum_{k \geq 5} N_k k(k-4))$, so triple points are the only degeneracy that can raise the ceiling, by at most one triangle each.*

4 A closed-form ceiling for K_{gen}

Lemma 8 (face budget). $|S| + C + Q + \sum_p \binom{k_p-1}{2} \leq \binom{n-1}{2}$.

Proof. $\mathcal{A}$ has exactly $\binom{n-1}{2} - Q - \sum_p \binom{k_p-1}{2}$ bounded faces. The interior of $T \in S$ contains at least $1 + c_T$ bounded faces, where c_T is the number of lines crossing it, and distinct members have disjoint interiors, so they consume disjoint face sets; summing gives $|S| + C$. $\square$

Theorem 9 (closed-form ceiling). $K_{\text{gen}}(n) \leq \lfloor (n-2)(5n-3)/12 \rfloor$ for all $n \geq 3$. In the combined inequality used, multiplicity 3 has coefficient exactly 0 and every $k \geq 4$ has coefficient $k(k-4) + 3\binom{k-1}{2} > 0$.

Proof. Add (2) to three times Lemma 8:

$$6|S| + 4C + 5Q + \sum_{k \geq 3} N_k \left(k(k-4) + 3\binom{k-1}{2} \right) \leq n(n-2) + 3\binom{n-1}{2} = \frac{(n-2)(5n-3)}{2}.$$

The bracket is 0, 9, 23, 42, ... for $k = 3, 4, 5, 6$, hence nonnegative. $\square$

Lemma 10 (reduction at zero slack). *If $\Delta = 0$ and $\mathcal{A}$ has no triple point, then $C = 0$, $Q = 0$, every multipoint has $k_p = 4$, and every member of S is a bounded triangular face; in particular $|S| \leq K(n)$ is attained by faces.*

Proof. (2) forces $C + 2Q + \sum_{k \geq 4} N_k k(k-4) \leq 0$ with all terms nonnegative. $\square$

Among known values, (2) is attained exactly at $n = 3, 5, 9, 15, 17$ — precisely the n where the Tamura value $n(n-2)/3$ is realised — and there all three quantities of Definition 1 coincide.

5 Certified small values in the strongest convention

The decision problem “is there an n -line arrangement with an admissible family of size T ?” is encoded in propositional logic over slope-sorted combinatorial data: variables for crossing versus parallel, concurrency of triples, the order of crossings along each line, the selection, and triangle/line crossing incidences, with the per-line inequalities of Lemma 4 and the budget of Lemma 8 as exact cardinality constraints. All degeneracies are free, so a verdict covers crossed and degenerate configurations simultaneously.

Theorem 11. *In the convention of Definition 1(iii),*

$$K_{\text{gen}}(4), \dots, K_{\text{gen}}(9) = 2, 5, 7, 11, 15, 21,$$

*and each value is certified: the instance $(n, K_{\text{gen}}(n) + 1)$ is unsatisfiable with a **drat-trim**-verified proof (Table 1), and the matching lower bounds are the classical constructions. Together with the prior eleven-cube cover for $n = 10$ this gives $K_{\text{gen}}(n)$ for all $n \leq 10$.*

n	$K_{\text{gen}}(n)$	refuted T	variables / clauses	core clauses	core lemmas	verify (s)
4	2	3	207 / 994	90	131	0.05
5	5	6	972 / 5,257	424	558	0.04
6	7	8	3,655 / 20,681	4,541	7,018	0.48
7	11	12	10,682 / 63,405	12,255	31,595	4.47
8	15	16	27,451 / 165,367	31,218	411,278	117.6
9	21	22	61,569 / 377,405	66,346	2,150,510	1060.3

Table 1: Certified initial segment. Every row is a `s VERIFIED drat-trim` run on a kissat proof of the refutation of $T = K_{\text{gen}}(n) + 1$; proofs range from 12kB to 328.5MB and the resolution counts from 582 to $8.7 \cdot 10^7$. Ledger: `scratch/kobon/ladder_certificates.json`.

These statements are strictly stronger than the corresponding classical ones, since they also exclude crossed families: e.g. $K_{\text{gen}}(8) = 15$ rules out the value $16 = \lfloor 8 \cdot 6/3 \rfloor$ left open for K_{gen} by the segment bound, and $K_{\text{gen}}(6) = 7$ rules out 8.

6 A verified lower certificate at $n = 14$

Theorem 12. $K_{\text{gen}}(14) \geq 54$, witnessed by an arrangement with exact rational coefficients.

Proof. From the SHA-256-pinned source file of [8] we verified: the 14 lines are pairwise non-proportional; the computed multipoint census is exactly two triple points sharing a line, with $Q = 0$, matching the file’s declared data; exactly 54 triples have interiors met by no other line, computed with two independent predicates, the second of which also rejects the case of a line through a vertex separating the opposite side; and the resulting family passes the audited exact-rational checker `verify_selection`, which validates non-degeneracy and pairwise interior-disjointness in homogeneous rational coordinates. The same verification succeeds for all fifteen published solutions. $\square$

7 Exact optimality of the known witnesses

For a *fixed* arrangement, the largest admissible family is the independence number of the overlap graph on non-degenerate triples, with edges joining pairs of intersecting interiors. We compute it exactly: interiors are compared in homogeneous rational arithmetic, and optimality is certified by SAT (a family of size K is a model, UNSAT at $K + 1$ certifies the optimum).

Theorem 13. *For each of the following the exact admissible optimum equals the published count, with certified optimality: the fifteen 14-line 54-triangle arrangements of [8]; two exact rational 53-triangle 14-line reconstructions; an 11-line 32-triangle arrangement; an 18-line 93-triangle reconstruction; and a 20-line 116-triangle reconstruction. In particular $K = K_{\text{gen}}$ on every known extremal arrangement.*

Consequently any improvement on the records must come from new arrangements, and by Corollary 7 any improvement past the Tamura value must use triple points — which is exactly the mechanism of Maiorana’s step from 53 to 54.

8 Equality-branch obstructions

When Δ is small the admissible degeneracy patterns form a short list, and saturation of Lemma 4 forces the multiplicities m_i of double-extreme vertices at consecutive direction classes to satisfy

$$m_{i-1} + m_i = 2s_i, \quad 0 \leq m_i \leq s_i s_{i+1}, \quad m_i \in \mathbb{Z}, \quad (3)$$

cyclically in the class sizes $(s_1, \dots, s_R)$. For odd R the rational solution is unique; for even R solvability requires $\sum_i (-1)^i 2s_i = 0$.

Theorem 14 ($n = 14$). *No arrangement of 14 lines with exactly three parallel pairs, or one parallel triad, and with no three lines concurrent admits an admissible family of 54 triangles.*

Proof. Paired layout: sizes are three 2's and eight 1's, $R = 11$ odd, and the unique solution of (3) violates an interval constraint for each of the $\binom{11}{3} = 165$ placements. Triad layout: one 3 and eleven 1's, $R = 12$ even, and the alternating sum is $\pm 4 \neq 0$ at each of the 12 placements. Both enumerations are certified by an exact integer program (pure standard library, no floating point). $\square$

Theorem 15 ($n = 18$). *No arrangement of 18 lines with exactly three parallel pairs, or one parallel triad, and with no three lines concurrent admits an admissible family of 94 triangles.*

Proof. Same mechanism with $R = 15$ and $R = 16$: all $\binom{15}{3} = 455$ paired placements violate an interval constraint (gap-parity classes $140 + 90 + 225$, 19 dihedral orbits with exact multiplicities, reflection quotients 231 and 9 all asserted by the certificate), and all 16 triad placements have alternating sum ± 4 . $\square$

Proposition 16 (the method stops at $n = 20$). *At $n = 20$, $T = 117$ we have $\Delta = 9$ and $2Q = 6$, leaving three units of slack, so (3) must be relaxed by a defect vector, and feasible defects exist: for the paired layout ($R = 17$) a defect of 2 at one doubled class with run lengths $(u, 0, 14 - u)$, e.g. $(6, 0, 8)$; for the triad layout ($R = 18$) a defect of 4 with all $m_i = 1$. Hence no argument using only (3) can exclude $T = 117$ at $n = 20$.*

9 The status of $K(14)$, and how to close it

Table 2 collects what is proved. The $n = 14$ row is the interesting one: the construction gives 54, (1) gives 55 for K , and Theorem 9 gives 67 for K_{gen} . Two decision instances close the gap in the strongest convention: $(14, 55)$, and — if that is unsatisfiable — nothing further is needed for K , while for K_{gen} the rungs $T = 56, \dots, 67$ remain, of which $T = 56$ is the first where Lemma 10 applies in the triple-point-free case. Both $(14, 55)$ and $(14, 56)$ instances (about $1.34\text{--}1.35 \cdot 10^6$ variables and $7.8 \cdot 10^6$ clauses) are running at the time of writing.

10 Open problems

1. **Kill the triple-point credit.** The $2a_\ell$ tokens in Lemma 4 are what allow $\sum_p k_p(k_p - 4) < 0$ and open the ceiling above Tamura's value. Conjecture: they are never simultaneously realisable, so $K_{\text{gen}}(n) \leq n(n - 2)/3$ with degeneracies. Evidence: equality in (2) occurs only at $N_3 = 0$ in the entire known census, and Theorem 13 shows crossings never help inside known witnesses.
2. **Decide** $(14, 55)$, closing $K(14)$ and hence the freshly recorded OEIS entry, and then the rungs up to 67 for K_{gen} .

n	construction	$K(n) \leq (1)$	$K_{\text{gen}}(n) \leq \text{Thm. 9}$	certified here
8	15	15	18	$K_{\text{gen}}(8) = 15$ (DRAT)
9	21	21	24	$K_{\text{gen}}(9) = 21$
10	25	26	31	$K_{\text{gen}}(10) = 25$ (cube cover, prior)
11	32	33	39	—
12	38	39	47	—
13	47	47	56	—
14	54 (verified)	55	67	$K_{\text{gen}}(14) \geq 54$
18	93	95	116	—
20	116	119	145	—

Table 2: Constructions are admissible families, hence lower bounds for all three quantities. Column three is the Clément–Bader draft bound for K (faces in arbitrary arrangements); its base value $\lfloor n(n-2)/3 \rfloor$ is peer-reviewed for non-simple arrangements [4], while the -1 for $n \equiv 0, 2 \pmod{6}$ rests on the draft [3]. Column four is Theorem 9, the only bound we know of that is proved for K_{gen} .

3. **Defect-tolerant obstructions.** Proposition 16 localises the missing ingredient: a constraint on where a defect may sit.
4. **A broad mod-6 refinement**, i.e. derive the -1 correction of (1) inside the framework of (2), making the K -bound rigorous for arbitrary arrangements and removing the reliance on an unpublished draft.
5. **Separation.** Is $K(n) < K_{\text{gen}}(n)$ for some n ? The first candidates are $n = 13$ ($T = 48$ requires a triple point and is excluded for K by (1)) and $n = 11$ ($T = 33$); both decision instances are running.

A Artefacts and replay

Release bundle `math/kobon/release/kobon-2026-08/` with `MANIFEST.sha256`; documentation hub `math/kobon/docs/`.

- `verification/maiorana.verify.py` — fail-closed certification of Theorem 12 (exit-0 transcript included).
- `verification/sweep.all.verify.py` — all fifteen witnesses.
- `verification/max_packing.py` — exact overlap graph and SAT optimality, used for Theorem 13.
- `scratch/kobon/broad_ladder.py` — ladder prober used for Theorem 11; instances `ladder_n8_t16.cnf`, `ladder_n9_t22.cnf` and the DRAT proofs.
- `proofs/appendix_independent_check.py`, `proofs/appendix_check_n18.py` — exact integer certificates for Theorems 14 and 15.
- `proofs/defect_lemma.md` — analysis behind Proposition 16.
- `scripts/engine.py` — exact rational geometry core and CNF builder.

References

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- [8] A. Maiorana, *Kobon triangles, $k = 14$* , dataset, https://github.com/rufio72/kobon_triangles_k14, commit e47c7cfd.
- [9] OEIS Foundation Inc., *Sequence A006066: Kobon triangles*, <https://oeis.org/A006066> (revision #227, 21 August 2026).
- [10] Companion technical document `math/kobon/paper_capacity.md`: full case analysis of Lemma 4, audit protocols, refutations of weaker penalty forms, and the complete experiment ledger.